MIL-C-48552(MU)
21 February 1975
SUPERSEDING
FA-PD-MI-2706, Rev. 0
31 January 1967

MILITARY SPECIFICATION
CASE, CARTRIDGE, 90MM, M19B1

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Case, Cartridge, 90MM, M19B1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect of date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-A-2550	-Ammunition and Special Weapons, General Specifications for

STANDARDS
Military

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-Quality Assurance Terms and Definitions
MIL-STD-1235	-Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167	-Ammunition Data Card
MIL-STD-1168	-Lot numbering of Ammunition
MIL-STD-1169	-Packaging, Packing, and Marking for shipment of inert Ammunition Components

FSC - 1315

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DRAWINGS

Armament Command

F7548080	-Case, Cartridge, 90MM, M19B1
F7548235	-Cartons, Ammunitions for Interplant Shipment of Steel Cartridge Cases, 37MM to 120MM, inclusive
LI-7548080	-Inspection Equipment List for Case, Cartridge 90MM, M19B1

(Copies of specification, standards, drawing and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI - B46.1	-Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

ASTM Method	-American Standards for testing and materials

(Application for copies should be addressed to the American Standards for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

ASTM Method E-8	-Tension testing of Metallic Materials
ASTM Method E-18	-Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM Method A317	-Macro-etch Testing and Inspection of Steel Forgings

(Application for copies should be addressed to the American Society for testing and materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 General.- The cartridge case shall comply with the requirements of the applicable drawings, referenced specifications and the following.

3.2 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.3 Mechanical properties.- The mechanical properties specified on the applicable drawing and referenced specifications shall be uniform throughout each section.

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3.3.1 Thermal treatment.- After heading operation, the entire cartridge case shall be subjected to a thermal temperature by heating above the upper critical temperature point, quenching and tempering to the desired properties.

3.4 Stress relief.- The cartridge case shall be adequately stress relieved at the times and temperatures suitable for development of desired properties.

3.5 Protective coating.- Protective coating as specified on the applicable drawing and referenced specifications shall apply to all surfaces of the cartridge case regardless of contour configuration.

3.6 Maximum profile.- The maximum profile of the cartridge case shall comply with the requirements of the applicable drawing.

3.7 Functional.- The cartridge case shall be subjected to a functional test in the following phases:

Excess Pressure Phase
Cold Temperature Phase
Excess, Excess Pressure Phase (for information only)

3.7.1 Obturation.- The cartridge case shall effectively seal the breech of the weapon against gases generated during firing. (info only)

3.7.2 Soundness.- After firing, the cartridge case shall be free of cracks or splits except as permitted by Table II.

3.7.3 Ejection.- After firing, the cartridge case shall be ejected by the service extractors when the weapon is operated in the normal manner (See 6.3).

3.7.4 Primer retention.- The cartridge case shall retain its primer during firing and ejection.

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following.

3.8.1 Metal defects.- The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches and porosity.

3.8.1.1 Flow of metal.- The flow of metal in the section of the cartridge case joining the head and the sidewall shall follow the direction of the contour of the interior curved surface without folds, cold shuts or roughness.

3.8.1.2 Visual standards.- Visual standards for metal defects will be established for each contract after the first three months production. However, establishment of visual standards will not waive the intent of the requirement for compliance with 3.8.1 and 3.8.1.1 (See 6.4)

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3.8.2 Burr.- No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile, or which might be injurious to personnel handling the cartridge case.

3.8.3 Foreign matter.- No cartridge shall contain chips, dirt, grease, oil, rust, corrosion, or other foreign matter during inspection or testing and prior to shipping.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance Terms and Definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance Terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First Article Sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 274 cartridge cases as follows:

254 coated
20 uncoated

The sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes, materials used, drawing or source of supply, such as to significantly affect product uniformity as determined by the responsible technical agency.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, 250 of the coated cases shall be forwarded to a designated proving ground as specified on the contract. Four of the coated cases and 20 uncoated cases shall be inspected for all the requirements of the drawings and specifications at a Government laboratory or such other facility as specified in the contact.

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4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions, test methods and procedures.

4.3.1 Lot.

4.3.1.1 Lot identification.- Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168.

4.3.1.2 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.3 Mechanical properties.

4.3.1.3.1 Hardness (test sample).- Twenty cartridge cases from the initial production sample and four cartridge cases from each production lot shall be subjected to this test. Failure of one or more cartridge cases to comply with the requirements of the applicable drawing shall be cause for rejection of the sample lot.

4.3.1.3.2 Hardness (test method).- This test shall be performed on locations specified by the applicable drawing in accordance with the procedures for Rockwell Hardness Testing prescribed by ASTM Method E-18.

4.3.1.3.3 Thermal treatment (test sample).- Each cartridge case shall be subjected to a hardness comparator test. Failure of the cartridge case to comply with the requirements of 3.3.1 or indications that the cartridge case failed to harden shall be cause for rejection.

4.3.1.3.4 Thermal treatment (test method).- This test shall be performed on a hardness comparator of non-destructive type approved by the technical agency.

4.3.1.3.5 Tension test (test sample).- Four cartridge cases from the initial production sample and four cartridge cases from each production lot shall be subjected to this test. The results shall be recorded for information only. The cartridge cases used for the hardness test may be used for this test.

4.3.1.3.6 Tension test (test method).- The test specimens shall be taken from the cartridge case in accordance with the standard positions and size of tensile specimen chart, Figure I. The yield strength, tensile strength and elongation shall be determined in accordance with the procedures prescribed for these tests by ASTM Method E-8.

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4.3.1.3.7 High stress expansion test (test sample).- Each cartridge case in the lot after heat treatment but prior to tapering shall be subjected to this test. Evidence of cracks in any portion of the case prior to or after tapering shall be cause for rejection of the case.

4.3.1.3.8 High stress expansion test (test method).- The cartridge case, after heat treatment shall be expanded diametrically .045 inches minimum while under pressure in a zone extending from 1½ inches above the flange to 9 inches above the flange, expansion from 1½ inches above the flange will decrease as the flange is approached, expansion at flange to be zero.

4.3.1.4 Macro-etch (test sample).- Four cartridge cases from the initial production sample and four cartridge cases from each production lot shall be subjected to this test to determine compliance with 3.8.1.1. Failure of one or more cartridge cases to comply shall be cause for rejection of the sample or lot. The cartridge cases used for the hardness test may be used for this test.

4.3.1.5 Macro-etch (test method).- The samples provided by 4.3.1.4 shall be cut in 90 degree sections along the axis through the primer hole. The specimen shall be etched in accordance with the procedures prescribed for this test by ASTM Method A317, and virtually examined to determine compliance with 3.8.1.1.

4.3.1.6 Protective coating (test samples).- The sample size and acceptance criteria shall be as specified in the applicable specification. Failure to comply with the requirements shall be cause for taking the corrective action prescribed therein.

4.3.1.7 Protective coating (test method).- The samples provided by 4.3.1.6 shall be tested using the procedures and equipment prescribed by the applicable specification.

4.3.1.8 Metal defects.- Prior to coating and after coating, each cartridge case shall be visually examined for metal defects. Failure of any cartridge case to comply with the requirements of 3.8.1 or established visual standards shall be cause for rejection of the cartridge case.

4.3.1.9 Maximum profile gage.- The cartridge case, after application of protective coating shall freely enter the Government certified maximum profile gage. The gate of the gage shall close without interference with the cartridge case head.

4.3.1.10 Functional.- The following functional tests will be performed by the Government.

4.3.1.11 First Article Sample.

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4.3.1.11.1 Initial production sample.- The quantities listed in Table I will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with the criteria set forth in Table II, except no retest will be permitted.

TABLE I

Test Phase	Quantity to be tested
Excess Pressure Phase	100
Cold Temperature Phase	50
Excess, Excess Pressure Phase (for information only)	100

4.3.1.12 Production lot sample.- One hundred cartridge cases from each first three (3) production lots will be ballistically tested in the excess pressure phase. After three (3) consecutive lots have been ballistically tested and accepted, the sample shall be reduced to fifty (50) cartridge cases after three (3) consecutive lots of which a sample quantity of fifty (50) have been ballistically tested and accepted, a sample quantity of twenty-five (25) will be used for the remainder of the contract for the excess pressure test. Whenever a ballistic failure occurs within a sample quantity of fifty (50) or twenty-five (25), the sample quantity will revert to one hundred (100) for subsequent lots until the above criteria for reduction is again satisfied Acceptance and rejection criteria shall be in accordance with Table II. The lot shall be considered eligible for retest, using double the number of sample provided that after firing, no cartridge case contains crack(s) or split(s) within one caliber of the head, and the rejection criteria of Table II has not been attained.

TABLE II

Characteristic	First Sample		Cumulative Sample	
	Accept	Reject	Accept	Reject
1. Obturation	(for information only)			
2. Splits or cracks within 1 caliber of the head	0	1	Not permitted	
3. Splits or cracks in sidewall more than 1 caliber from the head	1	3	2	3
4. Ejection	0	2	1	2
5. Primer retention	0	2	1	2

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4.3.1.13 Excess pressure phase (test method).- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees Fahrenheit (F) to plus 75 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.3.1.14 Cold temperature phase (test method).- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce a chamber pressure of 48,000 pounds per square inch (psi) to 50,000 psi piezo equivalent pressure when temperature conditioned at 70 degrees F. The complete round will be temperature conditioned at minus 35 degrees F., to minus 45 degrees F., and fired in a serviceable weapon for which round is standard. The weapon will be operated in the service manner.

4.3.1.15 Excess, excess pressure phase (test method).- The cartridge case will be assembled into a complete round in the same manner as required for service firing. The propellant charge will be such as to produce the maximum chamber pressure permitted by the proving ground. The complete round will be temperature conditioned at plus 65 degrees F., to plus 75 degrees F., and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.3.1.16 Inspection equipment.- Index of Inspection Equipment Lists Number LI-7548080 identifies the inspection equipment required to perform the examination and tests prescribed herein. Equipment design shall be in accordance with the code designation on the applicable LI.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105. Continuous sampling plans in accordance with standard MIL-STD-1235 may be used if approved by the procuring activity. An AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect shall be used. All nonconforming material shall be rejected. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination. A four power (4X) magnification may be used for visual examination.

4.3.2.1 Tool control dimensions.-Dimensions marked tool control shall be gaged during First Article Inspection and after each 250 items produced on each machine in operation and after any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimension and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Contractor will immediately institute corrective action whenever a defective tool control dimension is found.

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4.3.2.2 Case, Cartridge, Drawing F7548080 - Prior to coating

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
101.	True position of primer hole with body	Gage
102.	Diameter of flange, min.	Gage
103.	Diameter of shoulder under flange, min.	Gage
104.	Diameter of mouth	Gage
105.	Thickness of flange, min.	Gage
106.	Thickness of flange and shoulder, min.	Gage
107.	Thickness of head	Gage
108.	Thickness of wall at basic length near head	Gage
109.	Thickness of wall at basic length near midbody	Gage
110.	Thickness of wall at basic length near mouth	Gage
111.	Thickness of wall at mouth	Gage
112.	Contour of inside of head and sidewall not smooth	Visual
MINOR		
201.	Diameter of body under shoulder, min.	Gage
202.	Length to first basic diameter from head end, min.	Gage
203.	Length to second basic diameter from head end, min.	Gage
204.	Length to third basic diameter from head end, min.	Gage
205.	Length to fourth basic diameter from head end, min.	Gage
206.	Length total, min.	Gage
207.	Radii or chamfers, missing or incorrect	Visual (see 6.2)
208.	Surface finish, improper	Visual (see 6.2)
209.	Burr	Visual
210.	Foreign matter	Visual

4.3.2.3 Case, Cartridge, Drawing F7548080 - After coating

CRITICAL		
	None defined	
MAJOR		
101.	True position of primer hole counterbore with primer hole	Gage
102.	Diameter of primer hole	Gage

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4.3.2.3 (Cont'd)

Defect		Method of Inspection
103.	Depth of primer hole counterbore, min.	Gage
104.	Depth from front face of shoulder to bottom of primer hole counterbore, min.	Gage
105.	Taper in primer hole, incorrect	Gage
106.	Length total, max.	Gage
107.	Protective coating inadequate or damaged	Visual

MINOR

201.	Diameter of primer hole counterbore	Gage
202.	Dent or damaged	Visual
203.	Varnish in primer hole or primer hole counterbore	Visual
204.	Silicone compound missing from primer hole or primer hole counterbore	Visual
205.	Marking, missing, incorrect, or illegible	Visual

4.3.2.4 Carton, Packing - Drawing F7548235 - Prior to sealing

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Improper assembly	Visual
202.	Incorrect number of cartridge cases in carton	Visual
203.	Separator missing or damaged	Visual

4.3.2.5 Carton, Packing, Drawing F7548235 - After sealing

CRITICAL

None defined

MAJOR

None defined

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Carton damaged, cut or gouged through all layers	Visual
202. Carton improperly sealed	Visual
203. Contents loose	Manual
204. Marking missing, incorrect or illegible	Visual

4.3.3 Maximum profile (A Major Defect).- Each cartridge case shall be gaged in a Government certified maximum profile gage. Failure of the cartridge case to comply with the requirements of 3.6 shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A.- Not Applicable

5.1.2 Level B.- Not Applicable

5.1.3 Level C.- Cartridge cases shall be packed in accordance with the requirements of Drawing F7548235.

5.2 Labeling and marking.- Labeling and marking for shipment shall be in accordance with MIL-STD-1169.

5.3 Data card.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Prior to fabrication of an initial production sample, the contractor upon approval of the contracting officer, may make and submit for evaluation up to 5 process control lots. These process control lots shall consist of not less than 10 nor more than 25 cartridge cases each. The contractor shall furnish all information pertaining to the manufacture of the lots to the contracting officer. All costs for testing these process control lots will be borne by the Government.

6.2 Visual examination qualification.- When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

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6.3 Ejection.- The term "Ejection" shall mean that the cartridge case is thrown free of the breech of the weapon to permit loading of the subsequent round.

6.4 Visual standards.- Visual standards will be established at contractors plant in compliance with 3.8.1.2 or whenever directed by the Procuring Contracting Agency.

6.5 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

Custodian:

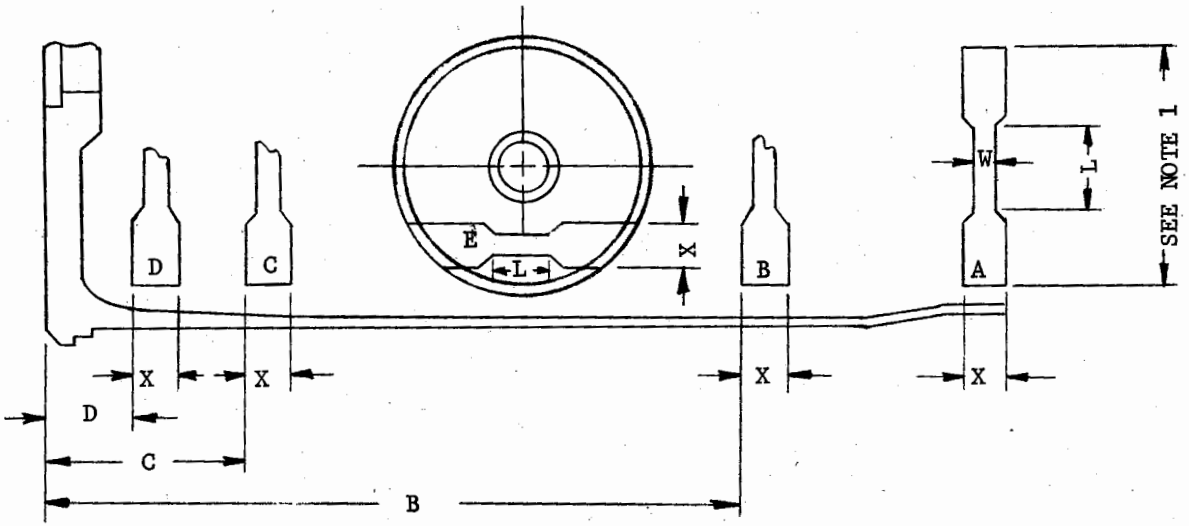
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Preparing Activity

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Project No. 1315 - A264

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SPECIMEN	X	W	L	LOCATION FROM HEAD
A	.40	.25	1.0	At Mouth
B	.40	.25	1.0	15.25
C	.40	.25	1.0	2.00
D	.40	.25	1.0	0.75
E	.40	.25	1.0	In Head

NOTES:

1 - Overall length of specimen to suit fixture in testing machine.

2 - Specimens C and D to be machined to uniform thickness parallel to inside surface.

3 - Specimen E to be taken from center of head midway between primer hole counterbore and edge of flange.

4 - Specimen E to be machined so as to eliminate effects of stamping on test.

5 - Yield strength in 0.2 percent offset.

6 - Percent elongation in 1.0 inch gage length.

STANDARD POSITIONS AND SIZES OF
TENSILE SPECIMENS FOR CASE, CARTRIDGE, 90MM, M19B1
FIGURE I

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